

Functional Requirements Document (FRD)

Clinical / Clinic Management System (CMS)

1. Introduction

- Purpose: Define the functional requirements for the Clinical Management System supporting patient registration, consultations, diagnostics, billing, and result management.

2. System Overview

- The CMS integrates workflows across departments including registration, consultation, laboratory, imaging, billing, teleconsultation, and reporting.
- Integrations include LIS, RIS, CIS, payment systems, email systems, and online patient portals.

3. User Roles

- Patient – registers for services and receives results.
- Customer Care Associate – registers patients and encodes requests.
- Physician – documents consultations and reviews results.
- Nurse/Clinic Staff – records vitals and assists consultation workflow.
- Laboratory Staff – processes specimens and validates results.
- Imaging Staff – records imaging results.
- Cashier – processes payments and generates receipts.
- System Administrator – manages users and security.

4. Patient Management Module

- Create patient records and assign unique IDs.
- Maintain demographics and contact details.
- Store medical history, consultations, and diagnostic results.
- Update patient information and enable record search.

5. Appointment and Scheduling Module

- Schedule physician consultations.
- Manage teleconsultation appointments.

- Schedule ECU and home services.
- Provide appointment dashboards and daily schedules.

6. Registration Module

- Patient check-in and queue number generation.
- Verify patient details.
- Register consultation and diagnostic services.
- Link requesting physicians and generate request forms.

7. Clinical Documentation Module

- Encode consultation details.
- Record complaints, medical history, and physician notes.
- Document physical examination findings.
- Generate prescriptions and consultation summaries.

8. Laboratory Management Module

- Generate specimen barcodes and labels.
- Track specimen receipt and routing.
- Encode laboratory results and validate results.
- Flag critical results and maintain QC records.

9. Imaging Management Module

- Register imaging procedures.
- Allow radiologists to encode interpretations.
- Store imaging reports and attach images.
- Validate imaging results prior to release.

10. Billing and Payment Module

- Record payment method (cash, HMO, corporate).
- Generate official receipts.
- Store payment records.

- Produce cashier summary reports.

11. Result Management Module

- Encode diagnostic results.
- Perform multi-level validation.
- Print or email results.
- Provide online result access.

12. Referral and Endorsement Module

- Enable branch-to-branch referrals.
- Refer patients for lab and imaging procedures.
- Track referral completion.

13. Monitoring and Reporting Module

- Monitor appointments and teleconsultations.
- Generate operational and workload reports.
- Maintain audit logs.

14. Security Requirements

- User authentication.
- Role-based access control.
- Protection of patient confidentiality.
- Activity logging.

15. Non-Functional Requirements

- Performance: support concurrent users.
- Reliability: high uptime and backups.
- Usability: intuitive interface.
- Scalability: support multiple branches.